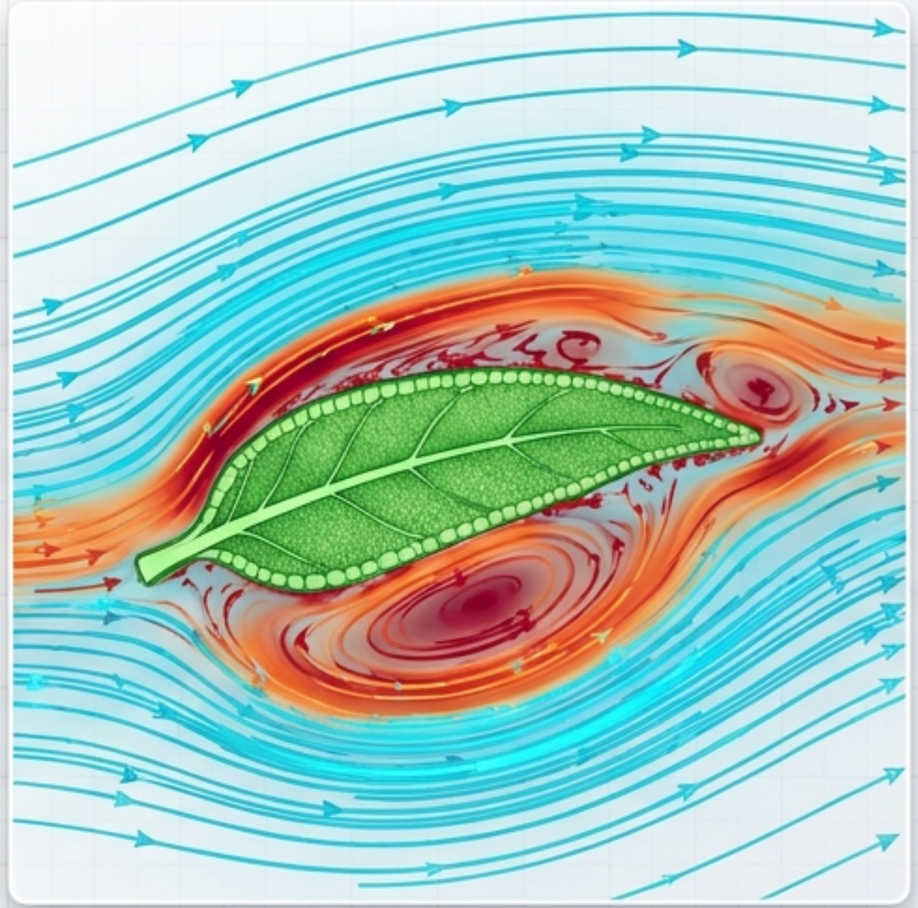
LunarLeaf-CFD

Validating gravity-dependent gas-exchange boundary layers for spaceflight botany.

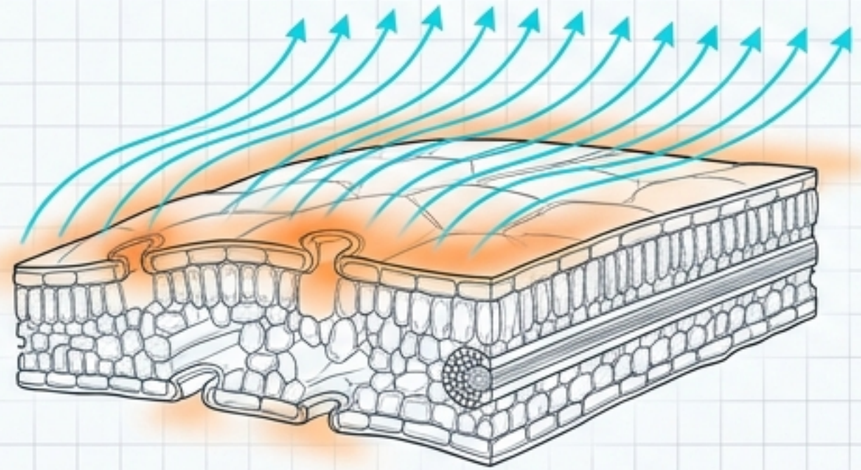


```
GRAVITY_VECTOR: [μg, Moon, Mars, 1g]
SCALES: [Leaf, Rosette, Canopy]
STATUS: Zenodo / npj Microgravity
```

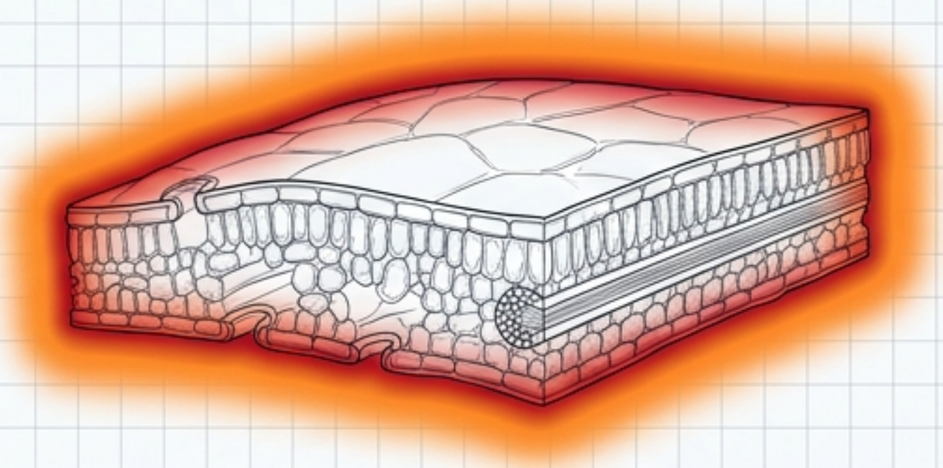
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In microgravity, the absence of buoyancy starves plant tissue.

Earth - 1g



Micrograv ity - µg



Buoyancy-Driven Convection

Photosynthesis warms and humidifies local air. Buoyancy forces the air to rise, continuously sweeping the boundary layer, refreshing CO_2 and removing O_2/H_2O.

The Stagnant Halo

Without gravity, natural convection collapses. Transport drops to pure molecular diffusion. Steep, toxic gradients of O_2, H_2O, and CO_2 depletion build up at the surface—triggering hypoxia and stress.

NotebookLM

Previous modeling tools solved cosmetics, not fluid physics.

Evolution Matrix	Aero-leaf-CFD	AstroBotany LEAF FDM	Nature's Aerodynamic Analyser	LunarLeaf-CFD
Strength	Great wizard UI.	Beautiful gravity-morphing shader.	Plain-language UI.	Browser-based GUI.
Fatal Gap / Breakthrough	Engine is Math.random(); results are noise	Boundary layer is a visual glow; static sliders, not a flow solve	Analysis is just LLM text; only ½ρv² is is real	Incompressible flow + multi-species advection-diffusion + Boussinesq buoyancy
Status Badges	[No Physics]	[No Physics]	[No Physics]	[Coupled Physics]

None of the prior prototypes coupled gravity to buoyancy. LunarLeaf-CFD exists to drop a professional CFD solver beneath a user-friendly interface.

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LunarLeaf-CFD: A user-friendly fluid dynamics workbench in the browser.

Zero Install:
Runs live in-browser (React 18 + Vite + TS).

Real-Time Sweeps:
Select gravity vectors (1g, Mars 0.38g, Moon 0.17g, µg) and watch fields evolve.



Full Diagnostics: Reports Grashof (Gr), Rayleigh (Ra), Sherwood (Sh), and Nusselt (Nu) numbers.

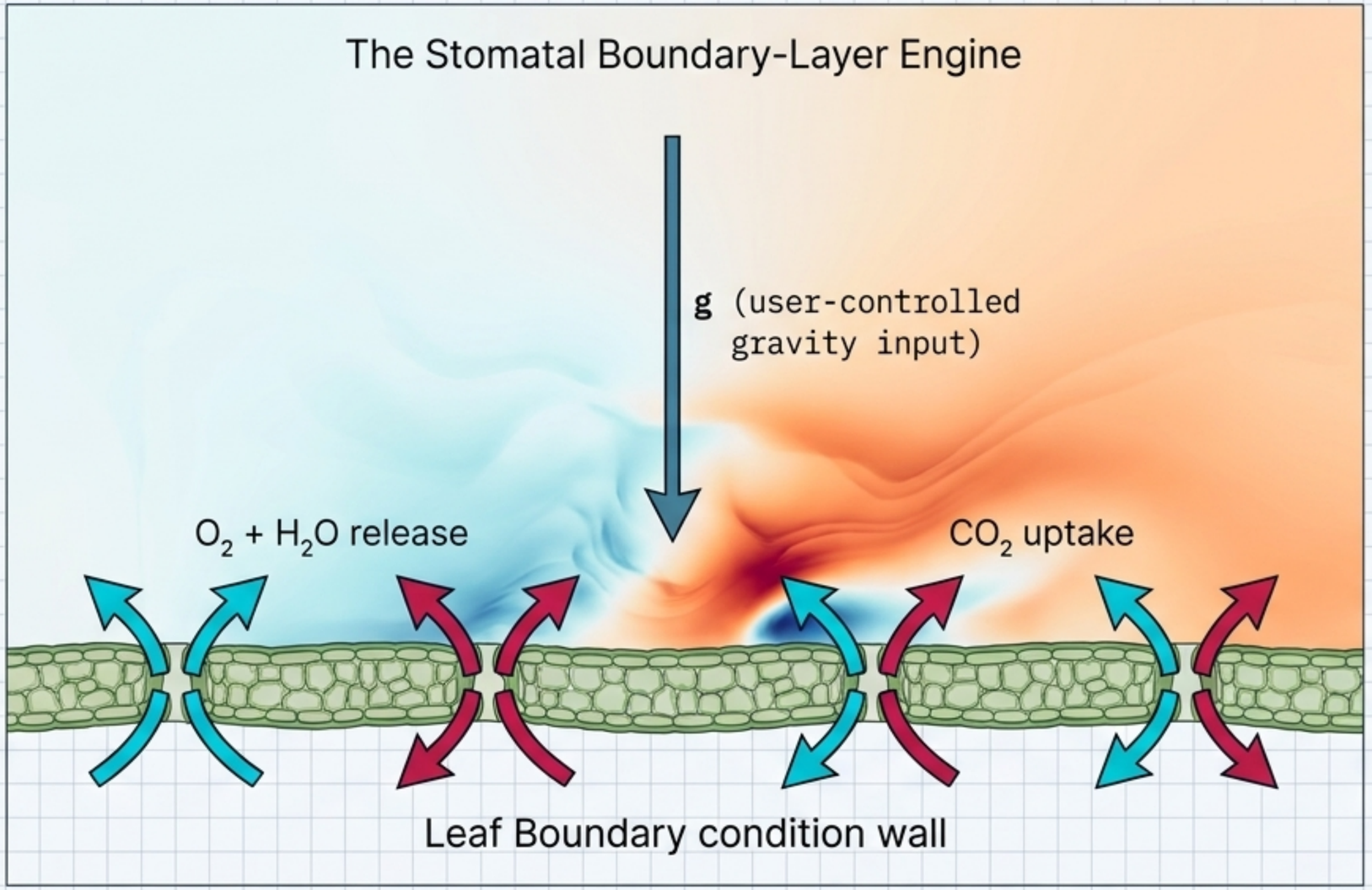
```
Gr = 1.2e6, Ra = 8.5e5,
Sh = 45.3, Nu = 22.1
```

Publication Ready:
Exports CSV, VTK, PNG, and MP4 data.

Packages professional FDM/CFD capabilities—voxelization, boundary conditions, field visualization—into a single web application

NotebookLM

The Physics Engine: Coupling biological flux to Boussinesq buoyancy.



Engine Specs

Core

D2Q9 Lattice-Boltzmann (BGK) incompressible flow solver

Transport

Multi-species scalar advection-diffusion (D2Q5)

Coupling

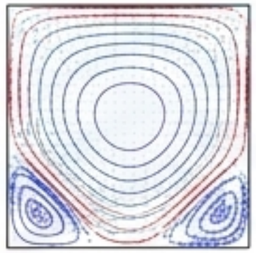
Boussinesq force directly tied to the user-defined gravity vector

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Validation: Passing textbook CFD benchmarks before modeling biology.

Gate 1: Lid-Driven Cavity (Re 100)

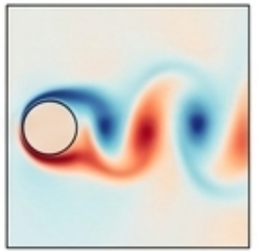
Proof: Centerline matches Ghia et al. (1982) with L2 = 0.0105.



[PASS]

Gate 2: Cylinder Flow (Re 100)

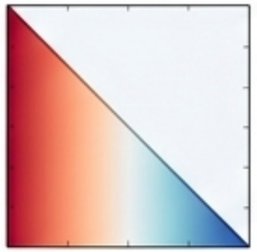
Proof: Captures Kármán vortex shedding; measured Strouhal number St = 0.192.



[PASS]

Gate 4: Pure Diffusion

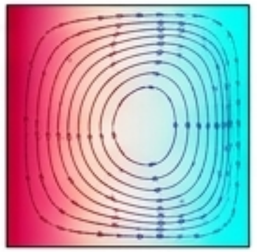
Proof: Matches ½·erfc analytic limit with L2 ≈ 0.0000.



[PASS]

Gate 3: Natural Convection (Ra 1e4)

Proof: Hot-wall Nusselt number matches de Vahl Davis (1983). Target: 2.238. Measured: Nu = 2.242 (0.18% error).



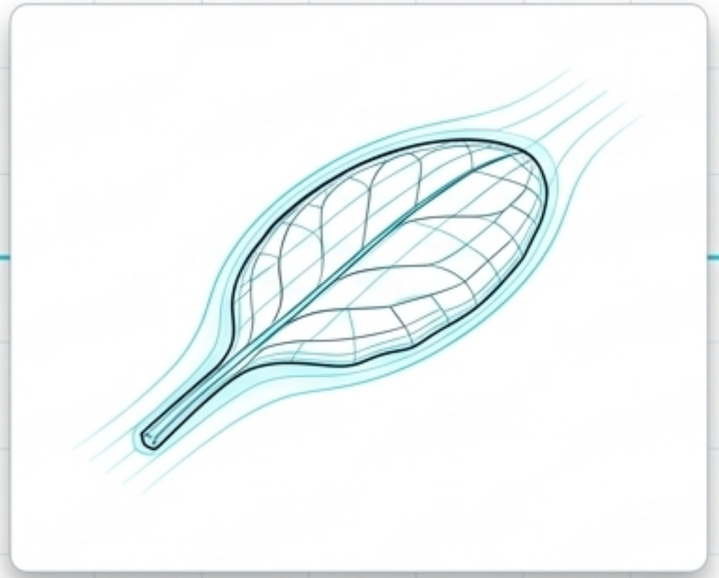
[PASS]

Gate 3 is decisive. It proves the engine correctly couples gravity to buoyant convection. Scaling **g** → 0 genuinely collapses the airflow.

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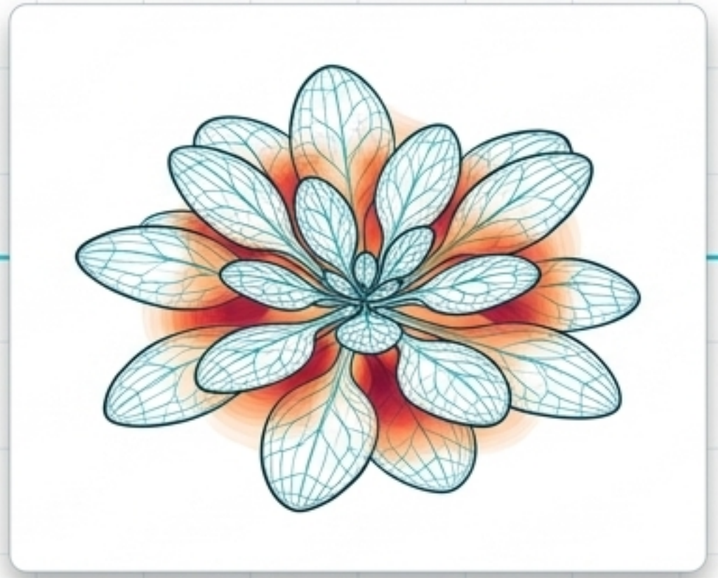
The Scale Cascade: Testing biological geometry at three levels.

1. Single Leaf



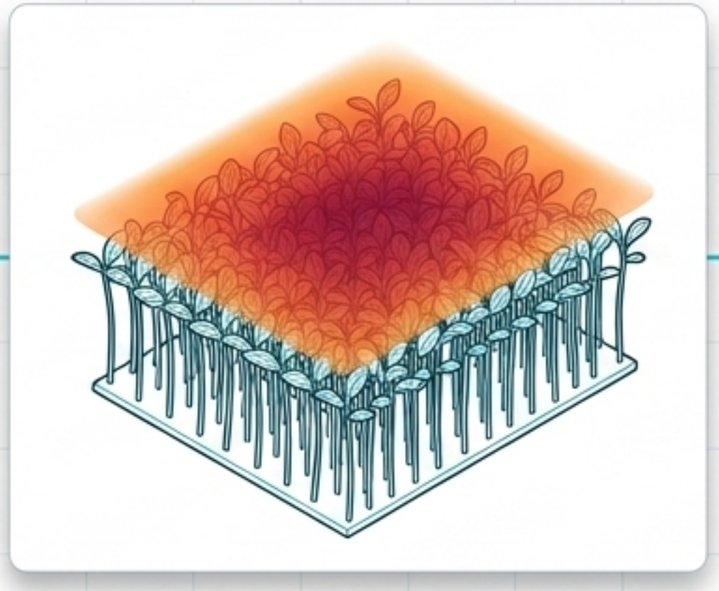
Boundary-layer physics on an isolated surface. Cleanest validation case.

2. Rosette



Leaf-to-leaf interaction. Introduces interior air-trapping and localized microclimates.

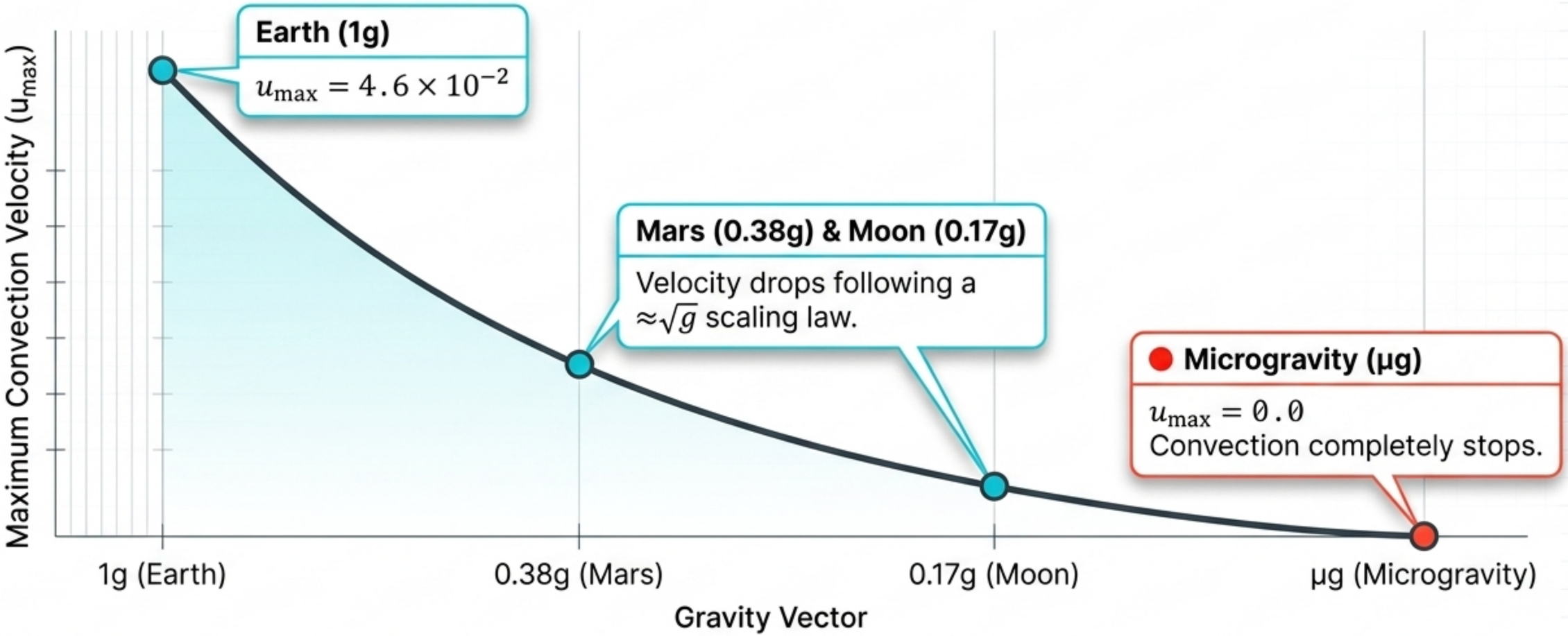
3. Microgreen Canopy



Emergent canopy boundary layer resulting in whole-stand stagnation.

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The Gravity Sweep: As gravity drops, natural convection dies.



Key Insight: Reducing gravity predictably starves the boundary layer of its mechanical sweeping force, leaving only molecular diffusion.

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The **Compounding Penalty**: Geometry and Microgravity amplify stagnation.

(Mean / Peak ΔC CO_2)

Scale	Gravity: 1g	Gravity: µg
Leaf	-0.161 / -0.246	-0.326 / -0.373
Rosette	-0.298 / -0.715	-0.554 / -1.033 (Deepest peak red—tlghtly enclosed pockets)
Canopy	-0.327 / -0.539	-0.445 / -0.602 (Deepest mean red—whole stand stagnation)

The **two effects compound**. A single leaf in **space** is **stressed**, but a **microgravity canopy** is the absolute **worst-case scenario** because **self-shading** traps already stagnant air.

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Mission Application: Modeling spaceflight growth hardware.

Spaceflight Hardware Matrix

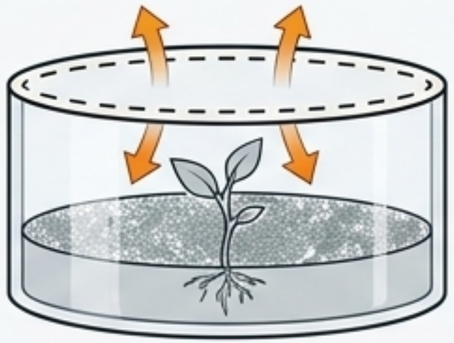
BRIC (Sealed)



Condition:
Fully closed volume.

Outcome:
CO_2 fixed in ~7 minutes.
O_2 hypoxia in ~6.5 days.
Closes the carbon loop; over a photoperiod, fixes only ~1% of Earth-equivalent carbon.

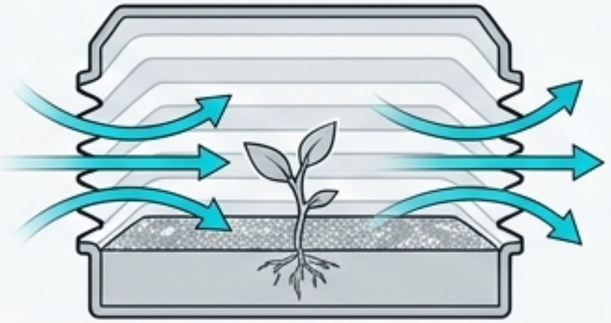
CARA (Micropore)



Condition:
Semi-permeable micropore tape membrane.

Outcome:
Baseline performance stabilizes; fixes ~90% of Earth-equivalent carbon.

VEGGIE (Vented)

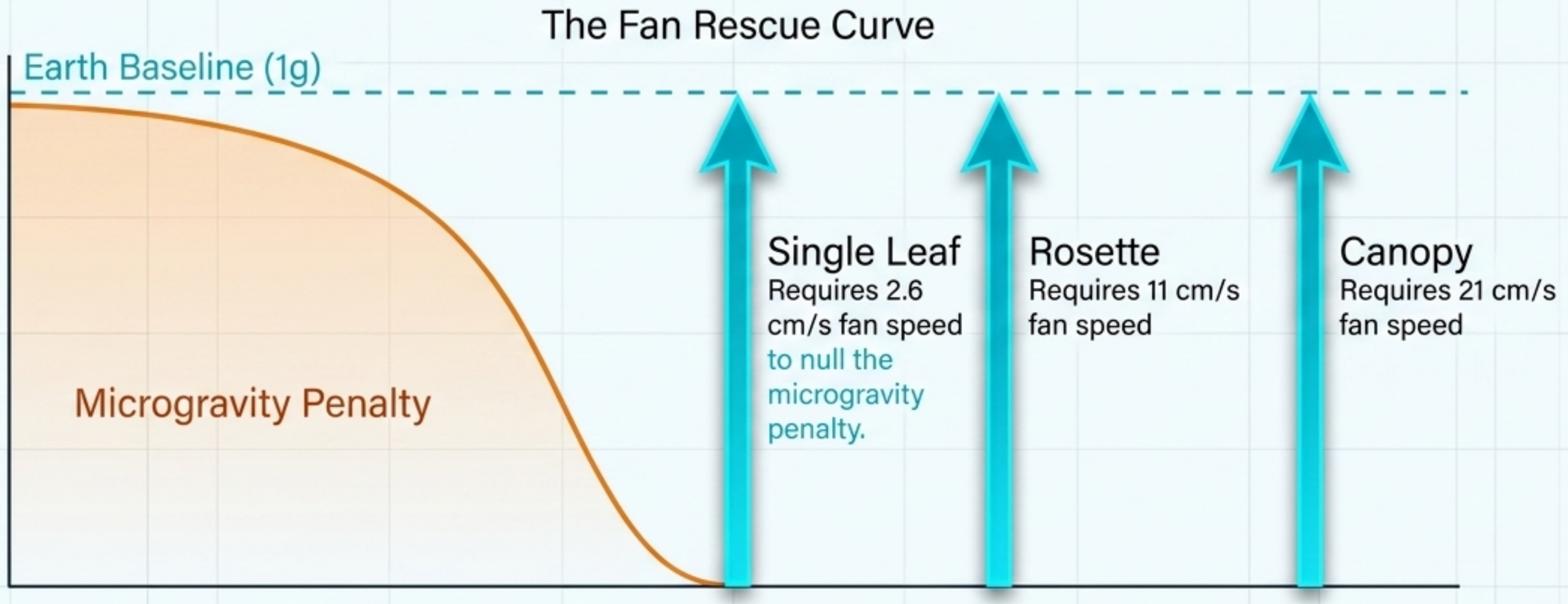


Condition:
Active airflow vents.

Outcome:
Fixes ~100% of Earth-equivalent carbon, highly dependent on forced air.

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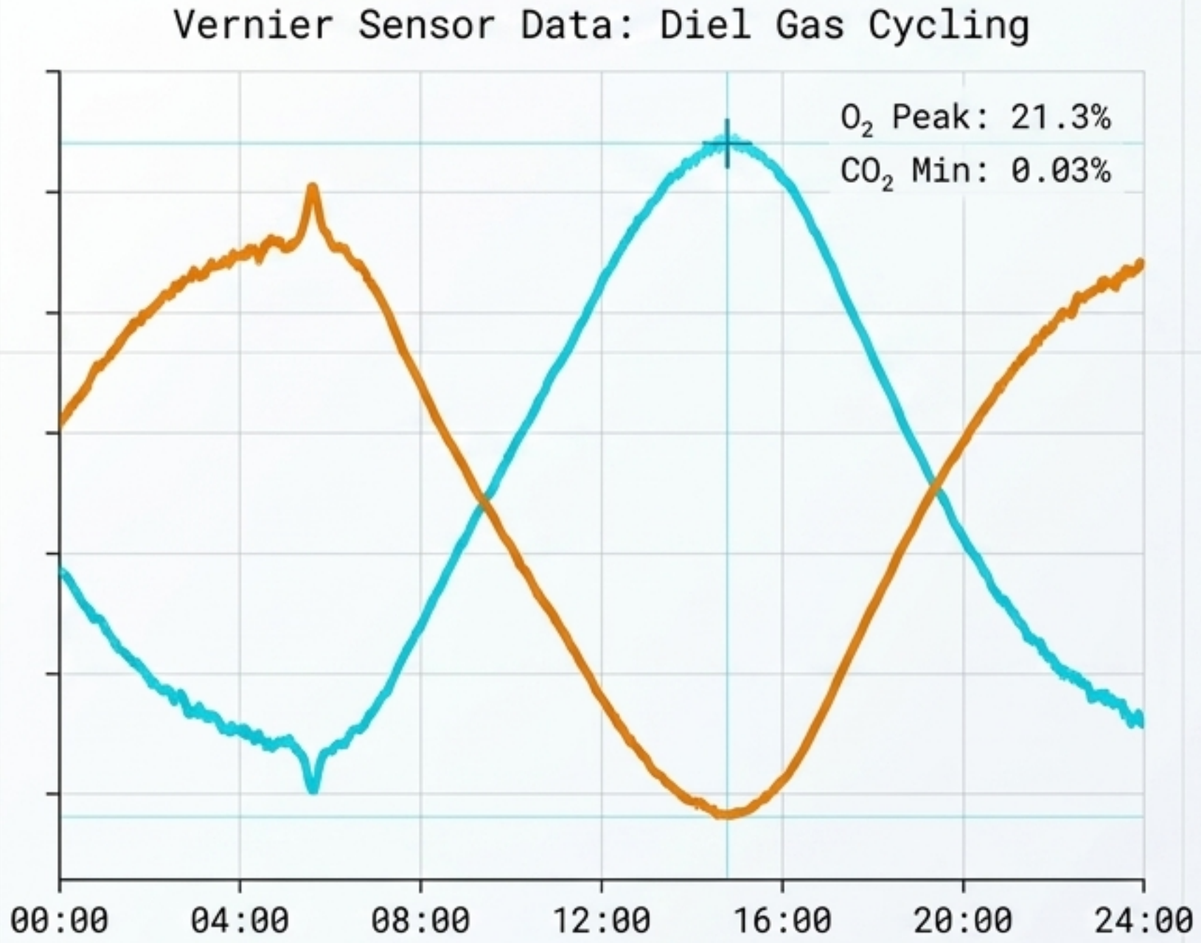
The Engineering Fix: Forced airflow required to rescue the boundary layer.



Takeaway: A single fan speed under-serves denser stands. Mission engineers must scale ventilation linearly with the density of the plant geometry.

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Real-World Anchoring: Validating against closed-chamber gas exchange.



Measured Flux Alignment

The model is firmly anchored to measured net assimilation data:
3.85 µmol CO_2 m^{-2} s^{-1} (biomass workbook).

Mass Conservation

Closed-chamber transport reproduced perfectly (mass conserved to 2×10^{-5}).

Physical Prediction

Predicts a surface CO_2 drawdown scaling from ≈4 ppm (Leaf, Earth) to ≈13 ppm (Rosette bulk, µg).

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Project Status: Open source, fully reproducible, and publication-ready.

The Code (GitHub)



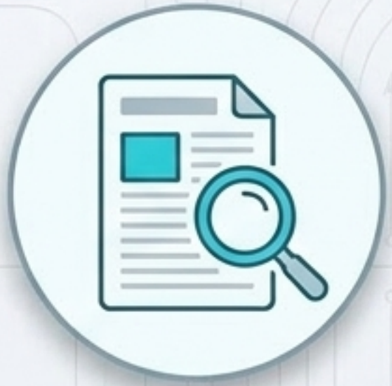
MIT Licensed.
Live browser demo deployed via GitHub pages.
No installation required.

The Archive (Zenodo)



Fully packaged release with `CITATION.cff`, `.zenodo.json`, and reference datasets.
Minted DOI for permanent academic archiving.

The Science (npj Microgravity)



Full manuscript draft assembled (CC-BY-4.0).
Self-contained LaTeX structure ready for npj Microgravity submission.

LunarLeaf-CFD: The fundamental physics of spaceflight botany, brought to the browser.

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